

AI Generated Syllabus

This syllabus is designed following the **AICTE** (All India Council for Technical Education) guidelines, tailored for a modern, industry-aligned curriculum. Since this is the 1st Semester of a B.Tech in Computer Science with an AI focus, the goal is to build a strong mathematical and logical foundation while introducing the "magic" of AI through hands-on coding.

Course Syllabus: Foundations of Artificial Intelligence & Computational Thinking

Course Code: CS101-AI

Course Name: B.Tech Computer Science (AI Specialization)

Semester: 1

Level: Beginner (Foundational)

Credits: 4 (L: 3, T: 1, P: 2)

Contact Hours: 60 Hours (Theory) + 30 Hours (Practical)

1. Course Overview

This course serves as the gateway to the world of Artificial Intelligence. Rather than jumping straight into complex neural networks, this course focuses on the "building blocks": logical reasoning, mathematical foundations (Linear Algebra and Probability), and foundational programming using Python. It bridges the gap between high-school mathematics and advanced machine learning, ensuring students understand *why* AI works, not just *how* to use it.

2. Course Objectives

- * To introduce the fundamental concepts and history of Artificial Intelligence.
- * To develop proficiency in Python programming specifically for data manipulation.
- * To build a strong mathematical intuition for AI (Linear Algebra, Calculus, and Statistics).
- * To introduce algorithmic thinking and problem-solving through search techniques.
- * To instill ethical awareness regarding AI deployment in society.

3. Detailed Units

Unit 1: Introduction to the AI Ecosystem (8 Hours)

* **The AI Landscape:** Definition of AI, Machine Learning (ML), and Deep Learning (DL); Turing Test and the history of AI.

* **Types of AI:** Narrow AI vs. General AI; Supervised, Unsupervised, and Reinforcement Learning overview.

* **AI in the Real World:** Computer Vision, NLP, Robotics, and Generative AI (LLMs) use cases.

* **Ethics & Bias:** Introduction to algorithmic bias, data privacy, and the social impact of automation.

Unit 2: Computational Thinking & Python for AI (12 Hours)

* **Python Fundamentals:** Data types, Control structures (if-else, loops), Functions, and Modules.

* **Data Structures for AI:** Lists, Tuples, Dictionaries, and Sets.

* **Algorithmic Logic:** Time and Space complexity (Big O notation) introduction; Flowcharts and Pseudocode.

* **Scientific Python Libraries:** Introduction to **NumPy** (Arrays), **Pandas** (DataFrames), and **Matplotlib** (Visualization).

Unit 3: Mathematical Foundations of AI - Part I (12 Hours)

* **Linear Algebra:** Scalars, Vectors, Matrices, and Tensors; Matrix multiplication; Identity and Inverse matrices; Transpose.

* **Calculus Essentials:** Derivatives and Gradients (the concept of "slope" in optimization); Introduction to Partial Derivatives.

* **Applications:** How matrices represent images and how gradients help in "learning."

Unit 4: Mathematical Foundations of AI - Part II (10 Hours)

* **Probability & Statistics:** Mean, Median, Mode, Variance, and Standard Deviation.

* **Probability Distributions:** Discrete vs. Continuous variables; Normal (Gaussian) Distribution.

* **Bayesian Thinking:** Conditional Probability and Bayes' Theorem (the logic behind spam filters).

* **Data Pre-processing:** Handling missing values, Outliers, and Data Normalization.

Unit 5: Search & Problem Solving (8 Hours)

* **State Space Search:** Defining problems as states and transitions.

* **Uninformed Search:** Breadth-First Search (BFS) and Depth-First Search (DFS).

* **Informed Search:** Introduction to Heuristics and Greedy Best-First Search.

* **Game Theory Basics:** Introduction to Minimax algorithm (The logic behind Chess/Tic-Tac-Toe AI).

4. Practical Assignments (Laboratory)

1. **Python Basics:** Implement a calculator and a student grading system using functions.
2. **Data Manipulation:** Use *Pandas* to analyze a CSV dataset (e.g., Titanic dataset or Weather data) and find averages/trends.
3. **Visualization:** Create histograms, scatter plots, and line graphs using *Matplotlib* to represent data distributions.
4. **Matrix Operations:** Implement matrix multiplication and transpose operations using *NumPy*.
5. **Search Algorithm:** Code a simple BFS/DFS algorithm to traverse a graph/maze.

5. Mini Project Ideas

* **Smart Weather Analyzer:** A Python script that reads historical weather data and predicts if it will rain based on simple probability.

* **AI Chatbot (Rule-Based):** A terminal-based chatbot that uses pattern matching to answer user queries.

* **Image-to-Matrix Converter:** A tool that loads an image and converts it into a NumPy array to show how pixels are represented numerically.

* **Tic-Tac-Toe Engine:** A game where a human plays against a computer using a basic Minimax algorithm.

6. Learning Outcomes

Upon successful completion of this course, students will be able to:

1. **Explain** the core concepts and terminology of the AI/ML domain.
2. **Write** efficient Python code to manipulate and visualize large datasets.
3. **Apply** linear algebra and probability concepts to solve computational problems.
4. **Implement** basic search algorithms to solve logic puzzles.
5. **Critically Evaluate** the ethical implications of AI technologies.

7. Recommended Software and Tools

- * **Language:** Python 3.x
- * **IDE/Environment:** Jupyter Notebooks, Google Colab (Recommended for zero-setup), or VS Code.
- * **Libraries:** NumPy, Pandas, Matplotlib, Seaborn.
- * **Version Control:** Basic Git/GitHub usage.

8. Suggested References and Books

- * **Textbooks:**
- * *Artificial Intelligence: A Modern Approach* by Stuart Russell and Peter Norvig (The "Gold Standard").
- * *Python for Data Analysis* by Wes McKinney.
- * *Mathematics for Machine Learning* by Marc Peter Deisenroth.
- * **Online Resources:**
- * **Coursera/DeepLearning.AI:** "AI For Everyone" (for conceptual clarity).
- * **Kaggle:** For datasets and hands-on practice.
- * **3Blue1Brown (YouTube):** "Essence of Linear Algebra" and "Essence of Calculus" series.